

5	(a)		$\rho = R \frac{A}{L} = R \frac{\pi(\frac{d}{2})^2}{L}$ $= 4.8 \left(\frac{\pi(\frac{0.23 \times 10^{-3}}{2})^2}{0.40} \right) = 4.9857 \times 10^{-7} \Omega \text{ m}$ $\frac{\Delta \rho}{\rho} = \frac{\Delta R}{R} + \frac{\Delta L}{L} + 2 \frac{\Delta d}{d}$ $= \frac{0.05}{4.80} + \frac{0.1}{40.0} + 2 \frac{0.01}{0.23} = 0.99873$ $\Delta \rho = 0.99873 \times 4.957 \times 10^{-7} = 4.979 \times 10^{-8}$ $= 5 \times 10^{-8} \Omega \text{ m}$ OR $\text{Maximum } \rho = 4.85 \left(\frac{\pi(\frac{0.24 \times 10^{-3}}{2})^2}{0.399} \right) = 5.4990 \times 10^{-7} \Omega \text{ m}$ $\text{Minimum } \rho = 4.75 \left(\frac{\pi(\frac{0.22 \times 10^{-3}}{2})^2}{0.401} \right) = 4.5028 \times 10^{-6} \Omega \text{ m}$ $\Delta \rho = \frac{1}{2}(5.4990 - 4.5028) \times 10^{-7}$ $= 4.980 \times 10^{-8}$ $= 5 \times 10^{-8} \Omega \text{ m}$	C1
				C1
				A1
				C1
				C1
				A1
(b)	(i)		Energy is dissipated in the internal resistance. Total energy is supplied by source is distributed to the external circuit and the internal resistance. Hence, electrical energy per unit charge available for external circuit is reduced.	M1
				A1
		(ii)	Calculation of R_x, $\frac{1}{4.8} = \frac{1}{12} + \frac{1}{R_x}$ $R_x = 8.0 \Omega$ Let V be the pd across the thermistor. $\text{Ratio} = \frac{\frac{V^2}{R_x}}{\frac{V^2}{12}} = \frac{8}{12} = \frac{12}{8} = 1.5$	C1
				B1
				A1
		(iii)	As temperature decreases resistance of thermistor X increases Total circuit resistance increases, total current decreases as EMF is constant. Power decreases.	B1
				M1
				A0

	(b)	(i)	Power of lamp exceed the power rating of the filament which is 60 W. The <u>filament had broken</u> and the <u>circuit has become open</u>. Note: When $V = 15.4 \text{ V}$, power of lamp = $15.4(5.9) = 90.86 \text{ W}$ $90.86/60 = 1.5$ times	B1
		(ii)	Resistance of the lamp is defined as the ratio of the pd across the lamp and the current passing through the lamp.	B1